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# Energy Storage Technology Selector

## Technical Documentation

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|                   |                                      |
|-------------------|--------------------------------------|
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| <b>Language</b>   | Vanilla JavaScript (ES Modules)      |
| <b>Deployment</b> | GitHub Pages (static, no build step) |
| <b>License</b>    | MIT                                  |

This tool provides indicative analysis only.  
Site-specific engineering studies are required before investment decisions.

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## 1 Overview

The **Energy Storage Technology Selector** is a browser-based interactive tool for comparing the lifetime economics of four energy storage technologies:

| Symbol        | Technology                  | Key | Color  |
|---------------|-----------------------------|-----|--------|
| Li-ion        | Lithium-ion battery         | li  | Green  |
| Vanadium Flow | Vanadium redox flow battery | va  | Blue   |
| Iron-Air      | Iron-air battery            | ia  | Amber  |
| Solid Oxide   | Solid oxide fuel cell       | so  | Purple |

The comparison is based on the **Levelized Cost of Storage (LCOS)** framework — the same methodology used by the U.S. Department of Energy’s Energy Storage Grand Challenge and Lazard’s LCOS reports. The tool runs entirely in the browser with no server, no database, and no build step.

### 1.1 Repo Structure

Listing 1: Repository file structure

```

1 energy-storage-selector/
2 |-- index.html           # HTML shell + all JS/CSS inlined (deployable
   standalone)
3 |-- css/
4 |   '-- styles.css       # All styling (dark theme, responsive)
5 |-- js/
6 |   |-- data.js         # Technology constants, app defaults, source
   citations
7 |   |-- model.js        # LCOS engine: npv(), calcTech(), fitScore()
8 |   |-- charts.js       # Chart.js rendering: cost breakdown + cash
   flow
9 |   |-- ui.js           # DOM manipulation, results rendering, wizard
   navigation
10 |   '-- main.js         # Entry point: event listeners, wires modules
   together
11 |-- README.md          # Methodology, equations, data sources
12 '-- LICENSE            # MIT

```

#### Note

`index.html` is fully self-contained — all CSS and JS from the modular files are inlined at build time. This means the app works when opened as a `file://` URL, in sandboxed iframes, and on GitHub Pages without any server configuration.

## 2 Scientific Methodology

### 2.1 Levelized Cost of Storage (LCOS)

LCOS is the energy-storage equivalent of the Levelized Cost of Energy (LCOE) used for power generation. It answers the question:

*“For every MWh this system ever delivers, what did it cost in today’s dollars?”*

The LCOS framework was formalised by Lazard (LCOS v1.0, 2015; current v8.0) and adopted as the official comparison methodology by the DOE Energy Storage Grand Challenge [1]. The academic foundation is the discounted cash flow (DCF) analysis from engineering economics [8].

### LCOS Formula

$$\text{LCOS} = \frac{C_{\text{capex}} + \sum_{t=1}^T \frac{C_{\text{replace},t} + C_{\text{O\&M},t} + C_{\text{charge},t}}{(1+r)^t} - \frac{V_{\text{residual}}}{(1+r)^T}}{\sum_{t=1}^T \frac{E_{\text{out},t}}{(1+r)^t}} \quad (1)$$

where:

- $T$  = project lifetime (years)
- $r$  = discount rate / WACC (decimal)
- $C_{\text{capex}}$  = upfront installed capital cost (\$)
- $C_{\text{replace},t}$  = replacement cost in year  $t$  (\$)
- $C_{\text{O\&M},t}$  = annual O&M cost in year  $t$  (\$)
- $C_{\text{charge},t}$  = charging electricity cost in year  $t$  (\$)
- $V_{\text{residual}}$  = end-of-life asset value (\$)
- $E_{\text{out},t}$  = energy discharged in year  $t$  (MWh)

## 2.2 Net Present Value (NPV) Helper

All future cashflows are discounted using standard NPV:

$$\text{NPV}(X) = \sum_{t=1}^T \frac{X_t}{(1+r)^t} \quad (2)$$

Implemented in `model.js` as `npv(cashflows, r)`.

## 2.3 Annual Energy Throughput

$$E_{\text{in}} = E_{\text{capacity}} \times \text{DoD} \times N_{\text{cycles}} \quad [\text{MWh/yr}] \quad (3)$$

$$E_{\text{out},t} = E_{\text{in}} \times \eta_{\text{RTE}} \times (1 - \delta)^{t-1} \quad [\text{MWh/yr}] \quad (4)$$

where  $\eta_{\text{RTE}}$  is the round-trip efficiency and  $\delta$  is the annual capacity degradation rate.

## 2.4 Vanadium Duration-Dependent Capex Model

Vanadium flow batteries decouple power cost (cell stack) from energy cost (electrolyte), unlike Li-ion where they are coupled. The effective system \$/kWh is:

$$c_{\text{stack}} = \frac{c_{\text{stack,kW}} \times P_{\text{MW}} \times 10^3}{E_{\text{MWh}} \times 10^3} \quad [$/kWh] \quad (5)$$

$$c_{\text{eff}} = \min(c_{\text{list}}, \max(0.60 c_{\text{list}}, (c_{\text{stack}} + c_{\text{electrolyte}}) \times f_{\text{scale}})) \quad (6)$$

with parameters:

- $c_{\text{stack,kW}} = \$500/\text{kW}$  (2025 commercial, Invinity/Largo Resources)
- $c_{\text{electrolyte}} = \$150/\text{kWh}$  (vanadium pentoxide + processing, Bushveld Minerals/PNNL 2023)
- $c_{\text{list}} = \$480/\text{kWh}$  (small-system list price, Mordor Intelligence 2025)

**Important**

The PNNL long-term R&D stack target is \$150/kW. This is *not* a current commercial price and must not be used for 2025 project economics. The real 2025 commercial stack cost is \$400–600/kW (Invinity / Largo Resources).

**2.4.1 Scale Discount Function**

Larger systems benefit from bulk electrolyte purchasing and shared balance-of-plant costs. The scale discount  $f_{\text{scale}}$  is:

| System size | Discount | $f_{\text{scale}}$ |
|-------------|----------|--------------------|
| < 10 MWh    | 0%       | 1.00               |
| 10–50 MWh   | 5%       | 0.95               |
| 50–200 MWh  | 12%      | 0.88               |
| 200–500 MWh | 20%      | 0.80               |
| > 500 MWh   | 28%      | 0.72               |

Sources: Lazard LCOS v8.0, PNNL 2023, Invinity project data 2024.

**2.5 Degradation Model**

Annual capacity fade reduces effective energy throughput over time:

$$E_{\text{out},t} = E_{\text{in}} \times \eta_{\text{RTE}} \times (1 - \delta)^{t-1} \quad (7)$$

| Technology     | $\delta$ (/yr) | Source                                  |
|----------------|----------------|-----------------------------------------|
| Li-ion         | 2.0%           | NREL Annual Technology Baseline 2024    |
| Vanadium Flow  | 0%             | Electrolyte ions fully recoverable      |
| Iron-Air       | 0.5%           | Conservative estimate, early-commercial |
| Solid Oxide FC | 0.5%           | DOE FCTO degradation targets            |

**2.6 Li-ion Replacement Cost**

Li-ion systems require periodic cell replacement. The replacement cost at year  $y$  is:

$$C_{\text{replace},y} = C_{\text{capex}} \times (1 - \delta)^y \times 0.85 \quad (8)$$

The 0.85 factor accounts for: (1) the degraded pack requiring less energy to replace, and (2) continued learning-curve cost reductions in Li-ion cells. Replacements are scheduled at years 8 and 16 (within a 20-year project life), consistent with NREL ATB 2024.

**2.7 Residual Value**

End-of-life asset value is discounted back to present value:

$$V_{\text{residual,PV}} = \frac{C_{\text{capex}} \times f_{\text{residual}}}{(1 + r)^T} \quad (9)$$

| Technology     | $f_{\text{residual}}$ | Rationale                                                |
|----------------|-----------------------|----------------------------------------------------------|
| Vanadium Flow  | 75%                   | Electrolyte retains commodity value; resale/reprocessing |
| Li-ion         | 5%                    | Scrap/salvage                                            |
| Iron-Air       | 5%                    | Scrap/salvage                                            |
| Solid Oxide FC | 5%                    | Scrap/salvage                                            |

#### Note

The 75% vanadium residual assumption is the single most sensitive parameter in the model. It assumes a functioning secondary electrolyte market (PNNL 2023, Invinity commercial practice). Reducing this to 10–20% narrows vanadium’s lifetime cost advantage significantly.

## 2.8 Safety Cost Multiplier (Li-ion)

Li-ion systems incur additional costs at elevated safety tiers due to NFPA 855 compliance requirements (fire suppression, separation distances, detection systems):

$$C_{\text{capex,adj}} = C_{\text{capex}} \times m_{\text{safety}} \quad (10)$$

| Safety tier | $m_{\text{safety}}$ | Context                            |
|-------------|---------------------|------------------------------------|
| Standard    | 1.00                | Industrial or remote grid site     |
| High        | 1.12                | Urban, near occupied buildings     |
| Critical    | 1.22                | Data center, hospital, dense urban |

Non-flammable chemistries (vanadium, iron-air, SOFC) carry no safety premium ( $m_{\text{safety}} = 1.00$  at all tiers). Source: FM Global / NFPA 855 compliance cost estimates.

## 2.9 Application Fit Score

Beyond financial metrics, the tool provides a qualitative fit score (0–100) for each technology:

$$S_{\text{fit}} = S_{\text{duration}} + S_{\text{app}} + S_{\text{cycle}} + S_{\text{safety}} + S_{\text{bonus}} \quad (11)$$

| Component            | Max pts | Logic                                                    |
|----------------------|---------|----------------------------------------------------------|
| Duration fit         | 25      | Full if $d \leq d_{\text{max}}$ ; linear penalty beyond  |
| Application affinity | 20      | Based on typical deployment patterns                     |
| Cycle life headroom  | 10      | $N_{\text{cycles}} \times T \leq N_{\text{cycle,rated}}$ |
| Safety priority      | 20      | Non-flammable gain when safety $\neq$ standard           |
| Bonus factors        | 25      | High-frequency cycling, long-duration sweet spot         |

## 3 Technology Parameters

All parameters are defined in [js/data.js](#) with inline source citations.

Table 1: Technology parameters used in the model (v1.2)

| Parameter           | Li-ion | Vanadium         | Iron-Air | SOFC   |
|---------------------|--------|------------------|----------|--------|
| Capex (\$/kWh)      | 110    | 480 <sup>a</sup> | 100      | 280    |
| RTE (%)             | 88     | 70               | 45       | 60     |
| Degradation (%/yr)  | 2.0    | 0                | 0.5      | 0.5    |
| Replacement years   | 8, 16  | —                | —        | —      |
| Max duration (h)    | 12     | 24               | 120      | 240    |
| Min viable dur. (h) | 1      | 8                | 24       | 12     |
| Cycle life          | 4,000  | $\infty$         | 10,000   | 10,000 |
| Residual value      | 5%     | 75%              | 5%       | 5%     |

<sup>a</sup> List price for small systems. Effective capex computed via Equation 6.

Table 2: O&M cost ranges ( $/MWh - yr$ )

| Technology     | Default | Range   | Source                                       |
|----------------|---------|---------|----------------------------------------------|
| Li-ion         | \$8     | \$6–12  | NREL Cost & Performance 2023                 |
| Vanadium Flow  | \$12    | \$10–18 | DOE/PNNL Energy Storage Grand Challenge 2020 |
| Iron-Air       | \$6     | \$4–8   | Form Energy targets                          |
| Solid Oxide FC | \$10    | \$8–14  | DOE FCTO                                     |

## 4 Code Architecture

### 4.1 Module Dependency Graph

```

data.js → model.js
data.js → ui.js
model.js → ui.js
charts.js → ui.js
data.js, ui.js, main.js (entry point)

```

Each module has a single responsibility:

- `data.js` — constants only; no logic
- `model.js` — pure functions; no DOM access
- `charts.js` — Chart.js rendering only
- `ui.js` — DOM manipulation only; imports from all above
- `main.js` — event listeners; exposes `goStep` and `runAnalysis` to `window`

### 4.2 `data.js` — Technology Constants

Exports four objects:

**TECHS** Per-technology parameters (see Table 1)

**APP\_DEFAULTS** Default cycles/duration per application type

**TECH\_PROFILES** Qualitative pros/cons per technology

**SOURCES** Full citation list rendered in the app footer

Listing 2: data.js — TECHS structure (excerpt)

```

1 export const TECHS = {
2   li: {
3     name: 'Li-ion',
4     color: '#22c55e',
5     capex: 110,           // $/kWh      BNEF Battery Price Survey 2025
6     rte: 0.88,           // AC-AC round-trip efficiency      BNEF/
7                           NREL
8     degRate: 0.02,       // 2%/yr      NREL ATB 2024
9     replaceAt: [8, 16],  // replacement years
10    maxDur: 12,           // max practical duration (h)
11    minViableDur: 1,      // minimum viable duration (h)
12    cyclLife: 4000,       // NREL: 3,000-5,000 cycles to 80% capacity
13    safetyMult: { standard: 1.00, high: 1.12, critical: 1.22 },
14  },
15  // va, ia, so follow the same structure ...
};

```

### 4.3 model.js — LCOS Engine

The scientific core of the application. All functions are pure (no side effects, no DOM access).

#### 4.3.1 Exported Functions

**npv(cashflows, r)** Net Present Value of an annual cashflow array. Implements Equation 2.

**scaleDiscount(energyMWh)** Returns the volume discount factor  $f_{\text{scale}}$  for vanadium systems.

**vanadiumEffectiveCapex(powerMW, energyMWh, listCapex)** Computes effective vanadium */kWh via Equation 6*.

**calcTech(key, p)** Main LCOS calculation for one technology. Returns a full result object (see Section 4.3.2).

**fitScore(key, p)** Qualitative application fit score (0–100).

#### 4.3.2 calcTech(key, p) — Return Object

Listing 3: calcTech() return object structure

```

1 {
2   key,           // technology key: 'li' | 'va' | 'ia' | 'so'
3   capex,         // total installed capex ($)
4   replaceCost,   // total nominal replacement cost ($)
5   replacePV,     // replacement cost present value ($)
6   elecCostPV,    // charging electricity cost PV ($)
7   rteLossPV,     // RTE loss cost PV (display only) ($)
8   omPV,          // O&M cost present value ($)
9   residualPV,    // end-of-life residual value PV ($)
10  totalCostPV,   // sum of all cost PVs minus residual ($)
11  revenuePV,     // total revenue present value ($)
12  lcos,          // levelized cost of storage ($/MWh)

```



```

13  netPV,                // net present value = revenuePV -
    totalCostPV ($)
14  cumulativeNet,       // array[lifetime]: cumulative cash position
    by year ($)
15  payback,             // simple payback year (null if not achieved)
16  annualRevenue,       // gross annual revenue year 1 ($/yr)
17  annualCost,          // annual charging + O&M cost ($/yr)
18  effectiveCapexPerKwh, // effective $/kWh after vanadium scaling
19  durCheck,            // boolean: duration <= technology maxDur
20  cyclCheck,           // boolean: cycles within rated cycle life
21  }

```

Listing 4: model.js — npv() and scaleDiscount() implementations

```

1  export function npv(cashflows, r) {
2    return cashflows.reduce(
3      (sum, cf, t) => sum + cf / Math.pow(1 + r, t + 1),
4      0
5    );
6  }
7
8  export function scaleDiscount(energyMWh) {
9    if (energyMWh < 10) return 1.00; // no discount
10   if (energyMWh < 50) return 0.95; // 5% discount
11   if (energyMWh < 200) return 0.88; // 12% discount
12   if (energyMWh < 500) return 0.80; // 20% discount
13   return 0.72;                // 28% discount (GWh-scale)
14 }
15
16 export function vanadiumEffectiveCapex(powerMW, energyMWh, listCapex)
17 {
18   const stackCostPerKwh = (500 * powerMW * 1000) / (energyMWh * 1000)
19   ;
20   const electrolytePerKwh = 150;
21   const sf = scaleDiscount(energyMWh);
22   const effective = (stackCostPerKwh + electrolytePerKwh) * sf;
23   return Math.min(listCapex, Math.max(listCapex * 0.60, effective));
24 }

```

#### 4.4 charts.js — Visualisation

Renders two Chart.js charts. Both functions destroy any existing instance before recreating to avoid canvas reuse bugs.

**renderCostChart(ranked, results)** Stacked bar chart: 20-year cumulative cost broken down by capex, replacement, O&M, RTE losses, charging electricity, and residual value credit. Technologies sorted by LCOS (best to worst).

**renderCashChart(ranked, results, lifetime)** Line chart: cumulative net cash position year-by-year over project lifetime. Winner drawn as a solid line; others dashed.

#### 4.5 ui.js — DOM & Results Rendering

**goStep(n)** Shows step *n*, hides others. Updates stepper indicator state (active / done). Scrolls to top.

- collectInputs()** Reads all form inputs and returns a structured parameter object **p**.
- runAnalysis()** Calls **collectInputs()**, runs **calcTech()** for all four technologies, sorts by LCOS, then calls all render functions.
- renderViabilityWarnings()** Shows amber warning block when a technology ranks highly but operates below its **minViableDur** threshold.
- renderRankTable()** Builds the comparison table with LCOS, capex, replacement, O&M, RTE loss cost, and application fit score per technology.

## 4.6 main.js — Entry Point

Wires all event listeners on DOMContentLoaded:

Listing 5: main.js — key responsibilities

```
1 import { APP_DEFAULTS, SOURCES } from './data.js';
2 import { goStep, runAnalysis } from './ui.js';
3
4 // Expose to window so HTML onclick attributes can call them
5 // (ES modules do not attach to window automatically)
6 window.goStep = goStep;
7 window.runAnalysis = runAnalysis;
8
9 // Application card clicks auto-populate cycles and duration
10 document.querySelectorAll('.option-card').forEach(card => {
11   card.addEventListener('click', () => { /* update defaults */ });
12 });
13
14 // Sliders live label update
15 bindSlider('duration', 'dur-val', v => `${v} h`);
16 bindSlider('dod', 'dod-val', v => `${v}%`);
17
18 // Suggested MWh hint
19 document.getElementById('power').addEventListener('input',
   updateEnergyHint);
```

### Note

The `window.goStep = goStep` assignment is critical. ES modules run in strict mode and do not attach exports to `window` automatically. Without this, the HTML `onclick` attributes cannot find the functions and all buttons are non-functional. This was the root cause of the button bug in earlier versions.

## 5 User Workflow

The app guides users through a 4-step wizard:

1. **Application** — Select use case (arbitrage, backup, frequency regulation, renewable firming, data center, utility BESS). Set storage duration (1–120 h slider) and safety priority.
2. **Project Scale** — Enter rated power (MW), energy capacity (MWh), project lifetime, annual cycles, and depth of discharge.
3. **Economics** — Enter charging price, selling price, optional capacity payment, discount rate, and per-technology O&M costs.

4. **Results** — Displays: winner badge, 6 key metrics, viability warnings, ranked comparison table, cost breakdown chart, cash flow chart, and qualitative fit assessment cards.

## 6 Minimum Viable Duration Constraints

Each technology has physical and economic constraints on minimum storage duration. The LCOS formula can rank a technology favourably even outside its viable range because it optimises cost-per-MWh without considering whether the system is physically operable at that duration.

| Technology     | Min dur. | Reason                                                                     |
|----------------|----------|----------------------------------------------------------------------------|
| Li-ion         | 1 h      | No practical lower limit for grid-scale applications                       |
| Vanadium Flow  | 8 h      | Stack cost makes < 8 h uncompetitive vs. Li-ion                            |
| Iron-Air       | 24 h     | Slow charge rate ( $\sim C/20$ ); designed for multi-day                   |
| Solid Oxide FC | 12 h     | High operating temperature ( $\sim 800^\circ\text{C}$ ); slow thermal ramp |

When a technology's minimum viable duration exceeds the user's requested duration, the app displays a  $\triangle$  **check duration** badge in the results table and a descriptive warning block above it.

## 7 Deployment

### 7.1 GitHub Pages (Recommended)

Listing 6: Deploy to GitHub Pages

```

1 git clone https://github.com/YOUR-USERNAME/energy-storage-selector
2 cd energy-storage-selector
3 git add .
4 git commit -m "Initial release v1.2"
5 git push origin main
6 # Settings -> Pages -> Deploy from branch -> main -> / (root) -> Save
7 # Live at: https://YOUR-USERNAME.github.io/energy-storage-selector

```

### 7.2 Local Testing

Listing 7: Local development server

```

1 cd energy-storage-selector
2 python3 -m http.server 8080
3 # Open: http://localhost:8080

```

#### Important

The modular `js/*.js` files use ES module `import/export` syntax and cannot be opened directly as `file://` URLs due to browser CORS restrictions. Always serve via HTTP for development.

**Exception:** `index.html` has all modules inlined as a plain `<script>` tag and works as a standalone file in all environments.

## 8 Model Limitations

- All costs are in **nominal USD**. No inflation adjustment is applied.
- The **75% vanadium residual value** assumption is the most sensitive parameter. It requires a functioning secondary electrolyte market. Reducing to 10–20% narrows vanadium’s advantage significantly.
- **Li-ion replacement** is modelled at a fixed schedule (years 8 and 16). Actual timing depends on real degradation, which varies with temperature, cycle depth, and C-rate.
- **SOFC RTE of 60%** is the electrical-only figure. With waste heat recovery the effective system RTE can reach ~70% (DOE FCTO target).
- **Iron-air capex** of \$100/kWh reflects early-commercial pricing. The long-term Form Energy target is < \$20/kWh, which is not yet a commercially available price.
- The model does not account for: grid interconnection costs, land costs, permitting timelines, revenue degradation from battery capacity fade, ancillary service revenue stacking, or project financing structure.

## 8 References

### 8 References

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